

Git & GitHub Concepts – Meet Hirpara Notes

1. Basic Commands

- `git init` → initialize local repo
- `git add .` → stage all changes
- `git commit -m "msg"` → commit staged changes
- `git push` → push commits to GitHub

2. Pushing to GitHub

- **First time repo** → create repo on GitHub → `git remote add origin <link>` → `git push -u origin main`
- **Next commits** → just `git push`

3. Git Checkout & Detached HEAD

- `git checkout <commit_id>` → go to old commit (read-only mode)
- Detached HEAD → edits/commits here are **temporary** unless you create a branch

4. Working with Old Commits

1. Checkout old commit
2. Create branch: `git checkout -b <branch_name>`
3. Make changes → commit
4. Return to main → merge: `git merge <branch_name>`

5. Merge Conflicts

- Git shows conflict markers:

```
<<<<<< HEAD
(current branch)
=====
(incoming branch)
>>>>>> branch-to-merge
```

- Resolve by choosing **Current / Incoming / Both**
- Stage changes → `git add`
- Commit merge → `git commit`

6. Tips

- `git log --oneline` → see commit IDs
- `git reflog` → find dangling commits
- Branches = safe timelines to test/fix changes
- Merge always affects **current branch**

Summary

- Detached HEAD lets you view old commits safely.
- To make changes, branch from old commit.
- Merge changes back to main to keep history intact.
- Always resolve conflicts carefully using Current/Incoming/Both options.
- Branches are key to safe experimentation and collaboration.